

Aiden Aquino

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EXPERIENCE

Undergraduate Researcher, JMU Mathematics & Statistics

Aug 2026 - Present

Mathematical Threat Modeling for IT/OT Convergence in Hyperscale Data Centers (CCI Grant)

- Threat model for the attack surface created when once-isolated OT and physical security systems converge onto shared IP networks, built with industry partner Red Trident, Inc.; encrypted tenant data makes sabotage, not exfiltration, the real risk.
- Led the Bayesian attack graph and probabilistic inference work and authored the prior-provenance methodology: ~40 cited rules under which a probability with no traceable source does not enter the model.
- Modeled how facility structure changes attack likelihood in ways vulnerability databases cannot express, pricing two devices with identical CVE scores as different attack edges: an unencrypted legacy Wiegand reader-to-panel channel versus OSDP v2.

Undergraduate Researcher, JMU Mathematics & Statistics

Jan 2026 - Present

AI-Powered Digital Twin for Water Distribution System Cybersecurity (CCI Grant)

- Digital twin of a water treatment plant built from real ICS attack telemetry (iTrust SWaT, BATADAL), behind an operator dashboard that recommends mitigations but never executes them.
- Built two unsupervised anomaly detectors on a shared feature pipeline: a scikit-learn Isolation Forest and a PyTorch LSTM autoencoder that reached 95.9% recall on confirmed attacks; rolling-window features raised Isolation Forest precision from 15.0% to 32.4%.
- Authored the ingestion pipeline, S3 dataset sync, and FastAPI dashboard; grew the test suite from zero to 569 tests as primary contributor and repository maintainer on a three-person team.

Computer Science Teaching Assistant - James Madison University

Lead Teaching Assistant

Aug 2026 - Present

- Supports a staff of roughly 25 across five core CS courses, from introductory programming through computer systems, as one of eight leads; mentors TAs on instruction technique and onboards new hires.
- Developed and delivered a TA training session on the VS Code debugger, preparing TAs to teach debugging as a skill rather than diagnose bugs on students' behalf.

Teaching Assistant

Aug 2025 - May 2026

- Assisted a 20-student section during class meetings and staffed evening lab hours serving five courses, helping students with assignments, debugging, and course concepts.

PROJECTS

Xavier - Real-Estate CRM (independent project, in daily production use)

May 2026 - Present

- Designed, built, and deployed a CRM for a practicing real estate agent: an ASP.NET Core API of 55 REST endpoints over Azure SQL with role-scoped JWT auth, served to a React frontend with GitHub Actions CI/CD.
- Built two-way Gmail sync as a background service polling Gmail's incremental history endpoint against a stored cursor, with RFC-compliant reply threading and automatic contact and deal matching.
- Migrated 2,184 cleaned contacts and 424 historical deals into Azure SQL with zero data loss, preserving 2,040 client notes dating to 2015 and verifying row counts table by table.

Box Programming Language (Interpreter)

Jan - May 2026

- Designed and implemented a custom language in Ruby: lexer, recursive descent parser with operator precedence, AST interpreter, and scoped runtime supporting variables, functions, and control flow, with source-position error reporting.

EDUCATION

James Madison University

Aug 2023 - May 2027 (Expected)

B.S. in Computer Science, Minor in Mathematics and Data Analytics

- GPA: 3.85 | Upsilon Pi Epsilon (International Honor Society for the Computing and Information Disciplines) | President's List (3 semesters), Dean's List (2 semesters)
- **Relevant Coursework:** Machine Learning, Algorithms & Data Structures, Software Engineering, Computer Systems I & II, Discrete Structures I & II, Programming Languages, Probability & Statistics, Linear Algebra

SKILLS

Languages: Python, Java, C# (proficient); C, SQL, Rust, Ruby, Haskell (familiar)

Tools & Cloud: Git, GitHub Actions, CI/CD, Azure (App Service, SQL, Static Web Apps), AWS (S3), pytest

Frameworks & Data: PyTorch, scikit-learn, Pandas, NumPy, FastAPI, ASP.NET Core, EF Core, SQL Server